

A Confidence-Gated Agent for Query–Organization Relevance: Routing Hard Cases from Fine-Tuned RuBERT to a Search-Augmented LLM

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Abstract

This report describes the project focused on linking news articles from Telegram channels to stock issuers using natural language processing (NLP) techniques: rule-based approach through regular expression for creating annotation and TF-IDF and Logistic Regression as a baseline model for multilabel classification. This project marks the beginning of a larger system aimed at delivering in-depth analytics to investors, helping them gain actionable insights and enhance their decision-making process in the financial market. Model evaluation relies on F1 score. The project repository, including code and details, is accessible here: <https://github.com/ChernayaAnastasia/rubert-llm-router>.

1 Introduction

Modern map and local-search services answer hundreds of millions of queries such as “*шиномонтаж 24 часа*” or “*кафе с животными в москве*”. For every query the system must decide which of the surrounding organizations are *relevant* and should be shown to the user. The quality of this decision directly shapes the product: irrelevant results erode user trust and increase abandonment, while relevant organizations that are wrongly filtered out lose visibility and customers. Because the task operates at the scale of an entire map service, even small improvements in relevance translate into a large absolute number of better answers, which makes the problem both practically important and economically significant.

Query–organization relevance is a short-text, asymmetric matching problem. On one side there is a very short, often underspecified user query; on the other, a structured organization listing containing a name, an address, a rubric (category), user reviews, and a price list. The difficulty is that relevance frequently depends on attributes that are *not literally present* in the listing text: whether

the place is open around the clock, whether it has some specific product or service or whether it is suitable for children or animals. Purely lexical methods cannot bridge this gap, and even a text-only neural classifier is limited to the information contained in the listing itself.

In this work we cast the task as binary classification (relevant vs. irrelevant) and propose a *confidence-gated hybrid cascade* instead of a single monolithic model. A compact fine-tuned transformer (rubert-tiny2) handles the bulk of the traffic at very low cost. A confidence threshold, chosen empirically through error analysis, routes only the uncertain cases to a more capable but more expensive large language model (LLM) agent. The agent can additionally call a web-search tool to retrieve exactly the kind of external, real-world knowledge (opening hours, services, price level) that the listing text lacks. Crucially, the decision of *whether* to add web search at all is not assumed in advance: we first build a taxonomy of the residual errors and enable search only because a measurable share of those errors provably require external facts.

This design distinguishes our approach from the three families of methods commonly applied to relevance problems. Compared with lexical baselines (TF-IDF, BM25), our system captures semantic similarity rather than surface word overlap. Compared with a single transformer cross-encoder, it selectively adds external knowledge and explicit reasoning, but only where the cheap model is uncertain. Compared with an LLM-only classifier, it keeps cost and latency low by resolving the easy majority of examples with a small model and invoking the LLM (and search) only on the hard tail. In short, the contribution of our work is not a new neural architecture but an error-analysis-driven routing policy that allocates expensive computation and external retrieval where, and only where, they measurably help.

Our contributions are the following:

- We prepare and characterize a binary query–organization relevance dataset based on Yandex Maps data, including a principled binarization of ambiguous (borderline) labels and a held-out out-of-distribution subset used as a sanity check.
- We establish two single-model baselines for the task: a TF-IDF + logistic regression classifier and a fine-tuned rubert-tiny2 transformer.
- We propose a confidence-gated hybrid agent (BERT → LLM → optional web search) whose routing threshold and search-vs.-no-search decision are both grounded in a quantitative error analysis.
- We evaluate the system in a cost-aware setting, reporting not only accuracy and macro-F1 but also the monetary cost of inference and the measured contribution of web search to the final quality.

1.1 Team

This project was prepared by **Anastasia Chernaya** from the EDA stage, preparation of the final dataset for training, implementation of the baselines

and the fine-tuned model, the error analysis, building of the hybrid agent, and making this report.

2 Dataset

The data is provided by Yandex and represents the results of relevance annotation performed by human assessors. The original data stored here and available only for the educational purposes.

2.0.1 Description of data

- 1.

- 2.

2.0.2

3 Stages of the Project

3.1

4 Metrics

4.1

5 Results

6 Conclusion

Next steps: